

CES 2103: DATA ANALYTICS

Syllabus Cheat Sheet, Formulas & Visualization Guide

A Comprehensive Review of Data Basics, Descriptive Statistics, Preprocessing & Hypotheses Testing

1 Data Basics & Variable Classification

The foundation of exploratory data analysis is the structured representation of real-world observations in a data table.

- **Observations (Rows):** Individual entities or events (e.g., patient records, transactions, car models) about which attributes are recorded.
- **Variables (Columns):** Specific characteristics or metrics measured across all observations (e.g., Age, Weight, Blood Group, Income).
- **Unique Identifier Variable:** A unique field (such as a customer ID) used solely for record linking. **Never used directly in data analysis** as it contains no analytical patterns.
- **Identical Value Variable (Constant):** A variable whose value is identical for every row (e.g., a "Calibration Setting"). While useless for analytical modeling, it is retained for data generation documentation or accuracy assessment.

Types of Variables & Measurement Scales

Variables are categorized based on their numeric precision and scale of measurement, which dictates the types of visualization and statistical models that can be applied.

1. **Nominal Scale (Categorization only):** Values represent distinct, unordered categories. No numeric meaning or arithmetic operations are possible.
 - *Examples:* Gender (Male/Female), Blood Group (A/B/AB/O), Eye Color, Marital Status, Country of Origin.
 - *Dichotomous/Binary Variable:* A nominal variable with exactly two possible outcomes, conventionally mapped to 0 and 1 in modeling (e.g., Diabetes: 1 = Yes, 0 = No).
2. **Ordinal Scale (Categorization + Rank Order):** Values can be naturally ordered or ranked, but the distance between consecutive ranks is unknown and cannot be compared.
 - *Examples:* Customer Satisfaction Rating (Poor, Fair, Good, Excellent), Academic Rank, Pain Level (Mild, Moderate, Severe).
3. **Interval Scale (Ordered + Equal Intervals, No True Zero):** Differences between values can be compared and share a consistent unit of measurement. However, there is no absolute/true zero point representing the total absence of the quality.
 - *Examples:* Temperature in Celsius or Fahrenheit (0° does not mean absence of heat), IQ Scores, Date of Birth. Ratios are not meaningful (10°C is not twice as hot as 5°C).

4. **Ratio Scale (Ordered + Equal Intervals + True Zero Point):** Numeric scales with a meaningful zero (absolute absence of the attribute). All arithmetic operations, including ratios, are fully valid.
- *Examples:* Weight (kg) (0 kg represents absolute absence of weight), Height (m), Income, Age, Bank Account Balance (120 mmHg is exactly twice the pressure of 60 mmHg).

2 Descriptive Statistics & Distribution Shapes

Descriptive statistics summarize variables by identifying their central point and characterizing their spread and variation.

Central Tendency Formulas

Mean (Sample Average)

The sum of all observations divided by the sample size.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Note: Highly sensitive to outliers. Used for Interval and Ratio scales.

Median

The middle value when the data is sorted from lowest to highest.

$$\text{Median} = \begin{cases} x_{\frac{n+1}{2}} & \text{if } n \text{ is odd} \\ \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2} & \text{if } n \text{ is even} \end{cases}$$

Note: Extremely robust to extreme outliers. Used for Ordinal, Interval, and Ratio scales.

Mode

The most frequently occurring value in the dataset.

$$\text{Mode} = \operatorname{argmax}_x \text{Frequency}(x)$$

Note: The only measure of central tendency applicable to nominal scale categories.

Variation and Spread Formulas

Range

The absolute difference between the maximum and minimum values.

$$\text{Range} = x_{\max} - x_{\min}$$

Interquartile Range (IQR)

The range spanning the middle 50% of the data, defined as the difference between the 3rd Quartile (Q_3 , 75% mark) and the 1st Quartile (Q_1 , 25% mark).

$$\text{IQR} = Q_3 - Q_1$$

Outlier Detection (Tukey's Fence)

Values falling outside the lower and upper whiskers of a box plot are flagged as statistical outliers.

$$\text{Lower Bound} = Q_1 - 1.5 \times \text{IQR} \quad \text{and} \quad \text{Upper Bound} = Q_3 + 1.5 \times \text{IQR}$$

Sample Variance (s^2)

The average of the squared deviations from the mean, adjusted for degrees of freedom ($n - 1$) to provide an unbiased estimate.

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Sample Standard Deviation (s)

The square root of the variance, expressed in the same units as the original data.

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Standard Score (Z-Score)

Represents how many standard deviations an individual observation (x_i) is away from the mean ($\bar{x}$).

$$z = \frac{x_i - \bar{x}}{s}$$

A Z-score of 0 is the mean; positive Z-scores are above average, and negative are below average.

Empirical Rule (68-95-99.7 Rule)

For an approximately normal distribution (bell-shaped curve):

- **Roughly 68%** of observations fall within $\bar{x} \pm 1s$ (i.e., $-1 \leq z \leq 1$).
- **Roughly 95%** of observations fall within $\bar{x} \pm 2s$ (i.e., $-2 \leq z \leq 2$). Exact mathematical calculation captures **95.45%** of the data.
- **Exact 95% Confidence Z-Value:** To capture exactly 95.00% of the data under a normal curve, boundaries must be pulled in slightly to $z = \pm 1.96$. Use 1.96 for precise formulas and 2 for mental estimates.

Distribution Shape: Skewness and Kurtosis

Beyond central tendency and variance, the overall shape of the frequency distribution provides critical insights for statistical testing.

Sample Skewness

Measures the asymmetry of the data distribution around the mean.

$$\text{skewness} = \left(\frac{\sqrt{n \times (n-1)}}{n-2} \right) \times \frac{\frac{1}{n} \times \sum_{i=1}^n (x_i - \bar{x})^3}{\left(\frac{1}{n} \times \sum_{i=1}^n (x_i - \bar{x})^2 \right)^{3/2}}$$

- **Symmetric (skewness ≈ 0):** Mean $\approx$ Median $\approx$ Mode (bell-shaped curve).
- **Positive/Right Skew (skewness > 0):** Tail is longer on the right side. Bulk of observations lean left. Mean $>$ Median $>$ Mode (e.g., Income distribution).
- **Negative/Left Skew (skewness < 0):** Tail is longer on the left side. Bulk of observations lean right. Mean $<$ Median $<$ Mode (e.g., Retirement age).

Sample Kurtosis

Characterizes the peakedness and thickness of the tails of the distribution.

$$\text{kurtosis} = \frac{n-1}{(n-2)(n-3)} \times \left[(n+1) \times \frac{\sum_{i=1}^n (x_i - \bar{x})^4 / n}{(\sum_{i=1}^n (x_i - \bar{x})^2 / n)^2} - 3 \right] + 6$$

- **Mesokurtic (kurtosis ≈ 3):** Normal distribution shape. (Excess kurtosis ≈ 0).
- **Leptokurtic (kurtosis > 3):** Sharp, pronounced peak with heavy tails. Prone to extreme outliers.
- **Platykurtic (kurtosis < 3):** Flat peak with thin, light tails. Values are spread out evenly.

Confidence Intervals (CI)

Calculates a range of values estimating a true population parameter (such as mean μ) with a specified level of confidence (e.g., 95% or 99%).

$$\text{Confidence Interval} = \bar{x} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \quad \text{where Standard Error (S.E.)} = \frac{\sigma}{\sqrt{n}}$$

Precondition for Small Samples: If the population standard deviation (σ) is unknown and the sample size is small ($n < 30$), a **t-distribution** with $df = n - 1$ must be used instead of the Z -distribution.

3 Data Preprocessing & Normalization

Data preparation is the most resource-intensive phase of an analytics project. It involves:

1. **Access & Combine Data Tables:** Ingesting and merging disparate datasets. Unit mismatches must be resolved (e.g., standardizing hospital records measuring weight in kilograms and pounds into a single uniform scale).
2. **Looking for Errors:** Correcting invalid non-numeric strings in numeric fields (e.g. replacing 'low' with a value) or removing out-of-range typing mistakes (e.g. 3,000 lbs recorded for human weight).
3. **Handling Missing Values & Duplicates:** Removing duplicate records and treating missing data through imputation or deletion.
4. **Converting Text to Numbers:**
 - Replacing ranked text with numbers (e.g., Low $\rightarrow$ 0, Medium $\rightarrow$ 1, High $\rightarrow$ 2).
 - Converting nominal categories into separate columns of 0 and 1, called **Dummy Variables**.
5. **Converting Continuous to Categorical (Binning / Discretization):** Helpful when numeric precision is unwarranted or to accommodate algorithms requiring categorical data (e.g., converting credit score to Poor, Good, Excellent).

Data Normalization Formulas

Normalization transforms numeric columns to a consistent, standardized scale to prevent variables with larger ranges (e.g., credit limit) from dominating variables with smaller ranges (e.g., age) during modeling.

Min-Max Normalization

Maps data values linearly to a newly specified range, typically $[NewMin, NewMax] = [0, 1]$.

$$x'_i = \frac{x_i - \text{OriginalMin}}{\text{OriginalMax} - \text{OriginalMin}} \times (NewMax - NewMin) + NewMin$$

Z-Score Normalization (Standardization)

Standardizes a column so that its mean (μ) is 0 and standard deviation (σ) is 1.

$$z = \frac{x_i - \mu}{\sigma}$$

Decimal Scaling Normalization

Normalizes values by shifting decimal points based on the absolute maximum value in the column.

$$x'_i = \frac{x_i}{10^n} \quad \text{where } n \text{ is the number of digits of the maximum absolute value}$$

4 Calculating Metrics & Statistical Tests of Relationships

Relationships between variables are evaluated using targeted statistical metrics determined by variable scales.

1. Pearson Correlation Coefficient (r) — Continuous vs. Continuous

Quantifies the strength and direction of the linear relationship between two continuous variables.

Pearson Correlation Formula

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y}$$

- **Range:** Scale from -1.0 (perfect negative correlation) to $+1.0$ (perfect positive correlation). 0.0 implies no linear relationship.
- **Direction:** Positive sign ($r > 0$) means both rise together. Negative ($r < 0$) means as one rises, the other falls.
- **Strength:** $r \geq 0.8$ is generally considered very strong; $r \approx 0.5$ is moderate.

2. Kendall's Tau (τ) — Ordinal vs. Ordinal

Measures the monotonic association (agreement in ranks) between two ranked or ordinal variables. It is robust to outliers and based on counting concordant pairs (n_c , moving in the same direction) and discordant pairs (n_d , moving in opposite directions).

Kendall's Tau-A (No Ties)

Used when data does not contain tied ranks.

$$\tau_A = \frac{n_c - n_d}{\frac{1}{2}n(n-1)}$$

Kendall's Tau-B (Adjusted for Ties)

Used when ties are present in the ranks (very common in ratings and Likert scales). Rescales the denominator based on ties in X only (t_x) and ties in Y only (t_y).

$$\tau_B = \frac{n_c - n_d}{\sqrt{(n_c + n_d + t_x)(n_c + n_d + t_y)}}$$

3. Two-Sample Independent t-Test — Continuous vs. Categorical (2 levels)

Tests whether the means of two independent groups are significantly different from each other.

- **Null Hypothesis (H_0):** $\mu_1 = \mu_2$ (no difference in population means).
- **Preconditions:** Independent samples, approximate normality of distributions, unknown population standard deviation.

Pooled-Variance Two-Sample t-Test (Equal Variances)

$$T = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{where } s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

The test statistic is compared to a t -distribution with degrees of freedom: $df = n_1 + n_2 - 2$.

Welch's Two-Sample t-Test (Unequal Variances)

Robust to variance differences, adjusting degrees of freedom.

$$T = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

4. One-Way Analysis of Variance (ANOVA) — Continuous vs. Categorical (> 2 levels)

Tests whether three or more independent group means are equal.

- **Null Hypothesis (H_0):** $\mu_1 = \mu_2 = \dots = \mu_k$ (all group means are equal).

- **Alternative Hypothesis (H_1):** At least one group mean is different.

One-Way ANOVA F-Statistic

Defined as the ratio of between-group variation (MSB) to within-group variation (MSW).

$$F = \frac{\text{Mean Square Between (MSB)}}{\text{Mean Square Within (MSW)}}$$

$$\text{MSB} = \frac{\sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2}{k - 1} \quad (df_1 = k - 1)$$

$$\text{MSW} = \frac{\sum_{i=1}^k (n_i - 1) s_i^2}{N - k} \quad (df_2 = N - k)$$

Where k is the number of groups, N is the total observations, and $n_i, \bar{x}_i, s_i^2$ are group statistics. A large F value indicates between-group variation dominates, warranting rejection of H_0 .

5. Chi-Square Test of Independence (χ^2) — Categorical vs. Categorical

Tests whether an association exists between two nominal or ordinal categorical variables represented in a contingency table.

- **Null Hypothesis (H_0):** The two variables are independent (no association).
- **Alternative Hypothesis (H_1):** The two variables are dependent (associated).

Chi-Square test (χ^2) and Expected Frequencies

The test statistic compares Observed (O_i) cell frequencies against Expected (E_i) cell frequencies:

$$\chi^2 = \sum_{i=1}^m \frac{(O_i - E_i)^2}{E_i}$$

$$E_{r,c} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total (N)}}$$

Degrees of freedom: $df = (r - 1) \times (c - 1)$ (where r is row count, c is column count).

5 Data Visualization Selection Guide

Selecting the correct visualization depends on the variable scales and the specific relationship being investigated.

Table 1: Systematic Visualization Matrix

Chart Type	Independent (X)	Dependent (Y)	Analytical Condition / Use Case
Bar Chart	Nominal/Ordinal	Continuous/Discrete	Compare counts/sums or means across
Histogram	Continuous (Binned)	Frequency (Count)	Display shape, symmetry, and spread of
Box Plot	Nominal/Ordinal	Continuous (Ratio/Int)	Show spread, quartiles, symmetry, and o
Scatter Plot	Continuous	Continuous	Identify linear/nonlinear correlation and o
Line Graph	Continuous (Time)	Continuous	Track progress, trends, or forecasting ove
Stacked Bar	Categorical (Group)	Category Composition	Visualize sub-components or relative cor
Pie Chart	Nominal/Ordinal	Relative Proportion	Show static share of parts to a whole (be
Bubble Graph	Continuous (X-axis)	Continuous (Y-axis)	Plot up to 4 dimensions (size = 3rd, color
Dials	Continuous	Speedometer Zones	Instant view of a single KPI range (Red-Y
Pictograph	Nominal/Ordinal	Icon Counts	Highly interpretable count display using s
Contingency Table	Nominal/Ordinal	Nominal/Ordinal	Summarize counts of categorical combin

Exploratory Visual Examples

Below are styled examples of core exploratory graphs with their mapping conditions.

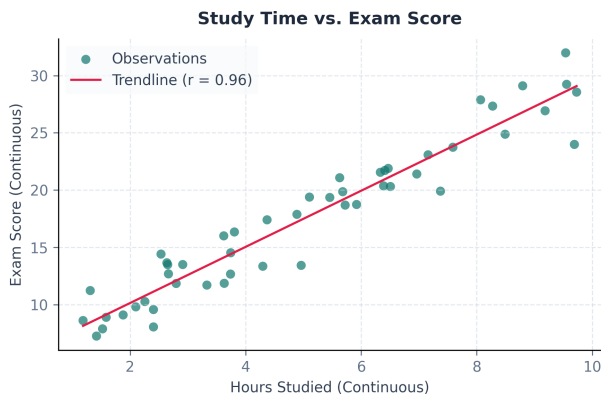


Figure 1: Scatterplot with a linear trend line (Continuous X vs. Continuous Y).

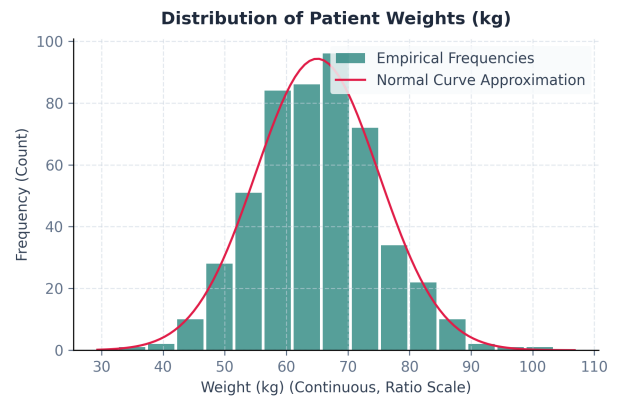


Figure 2: Histogram of weights with a normal density curve overlay.

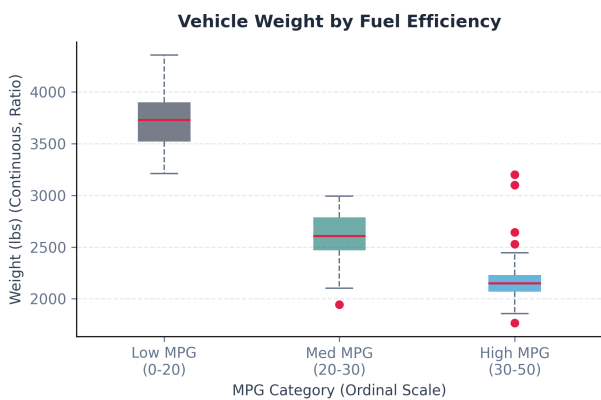


Figure 3: Grouped box plot comparing weights across ordinal MPG categories.

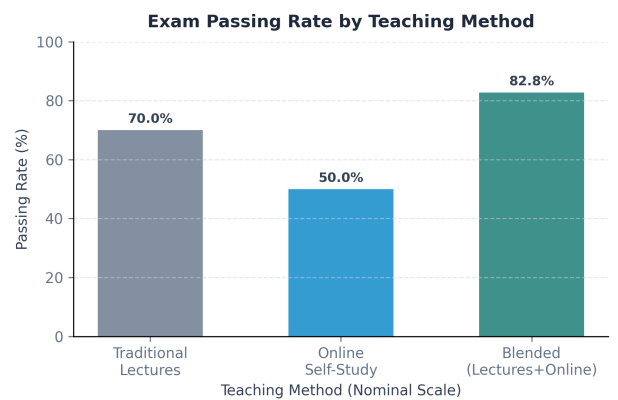


Figure 4: Categorical Bar Chart displaying mean pass rates across nominal methods.